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Synthesis of Electron-Deficient Heteroaromatic 1,3-Substituted Cyclobutyls via Zinc Insertion/Negishi Coupling Sequence under Batch and Automated Flow Conditions

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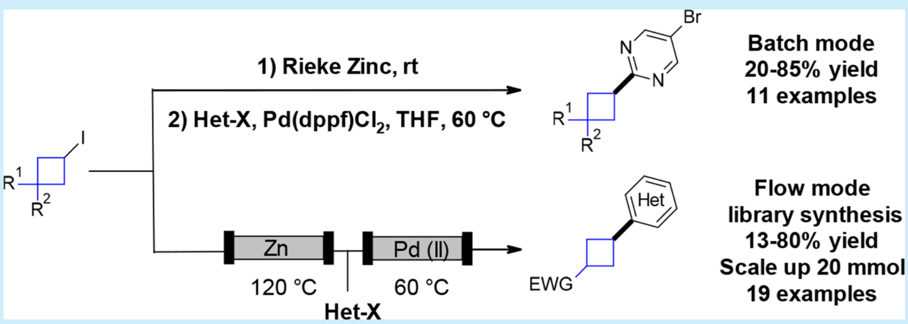
Supporting Information



ABSTRACT: Synthesis of 1,3-substituted cyclobutyls enabled by zinc insertion into functionalized iodocyclobutyl derivatives followed by Negishi coupling with halo-heteroaromatics is reported. Two distinct sets of conditions were developed; the first involved a two-step batch protocol using activated Rieke zinc, and the second involved a multistep continuous flow process. Both methods showed complementarity and allowed for rapid access to these medicinally relevant motifs, the possibility of scaling up, and automation for library synthesis.

Substituted cyclobutanes have generated significant interest within drug discovery and are frequently incorporated into many and varied drug candidates (Figure 1).[1] Due to the unique steric properties of the core structure, this small ring system has been extensively used to explore the chemical space.

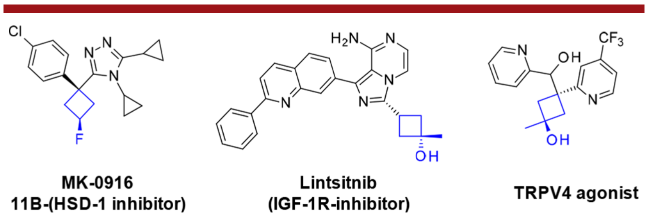


Figure 1. Medicinally relevant compounds containing 1,3-substituted cyclobutanes.

Over the past 30 years, Pd-catalyzed Negishi cross- coupling has been a versatile tool to create C−C bonds, notably in the presence of sensible functionalities.[2] Among a myriad of useful methodologies to couple $C_{sp}{}^3$−$C_{sp}{}^2$ under Negishi conditions,[3] Knochel reported a highly diastereoselective Pd-catalyzed cross-coupling reaction between functionalized cyclohexylzinc reagent and aryl iodide (Figure 2, eq 1).[4] In this two-step

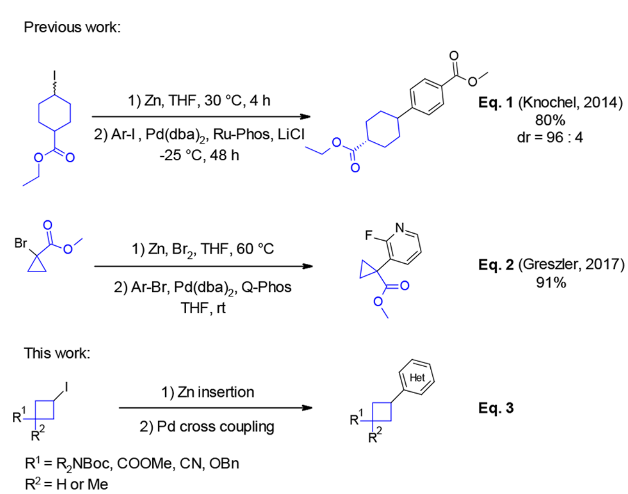


Figure 2. Negishi cross-couplings with functionalized cycloalkyls.

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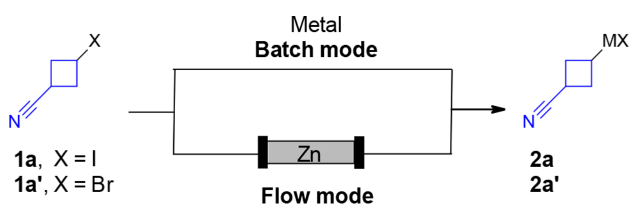
process, the organozinc is preformed via the insertion of preactivated zinc metal[5] and then engaged in the cross-coupling. This sequence has also been applied successfully to introduce the cyclopropyl moiety, taking advantage of the high reactivity of the corresponding Reformatski reagent (Figure 2, eq 2).[6] Inspired by these two methodologies and considering the potential for the emergence of flow chemistry to generate organometallic reagents,[7] we identified an opportunity to develop a general method to synthesize 1,3-disubstituted heteroaryl cyclobutanes (Figure 2, eq 3). Indeed, rapid access to this attractive motif is scarce and limited to α-arylation of cyclobutyl carbonyl derivatives[8] and direct addition of organolithium or organomagnesium reagents to cyclobutanone derivatives.[9] Isolated examples of cross-coupling reactions between cyclobutyl iodides and arylboronic acids have also been achieved.[10] Herein, we report a batch and a flow cross-coupling procedure between halo-heteroaromatics and various functionalized cyclobutylzinc. We were also able to achieve automation of a multistep flow process.

We started our investigation with the insertion of magnesium to 3-iodo- and 3-bromocyclobutane carbonitrile **1a** and **1a′**. Interestingly, no metal insertion was observed even after chemical activation of the magnesium turnings by TMSCl and dibromoethane[11] (Table 1, entries 1 and 2). Next, using 3 equiv

Table 1. Optimized Metal Insertion



entry	method	X	metal	activation	temp (°C)/time (min)	conv (%)[a]
1	batch	Br	Mg	TMSCl, $Br_2C_2H_6$	60/60	0
2	batch	I	Mg	TMSCl, $Br_2C_2H_6$	60/60	0
3	batch	Br	Zn[b]	HCl	25/60	62
4	batch	I	Zn[b]	HCl	25/60	85
5	batch	Br	Zn	Rieke[c]	25/60	56
6	batch	I	Zn	Rieke[c]	25/20	100
7	flow	I	Zn	HCl	80/6	10
8	flow	I	Zn	HCl	120/6	100
9	flow	Br	Zn	HCl	120/6	5

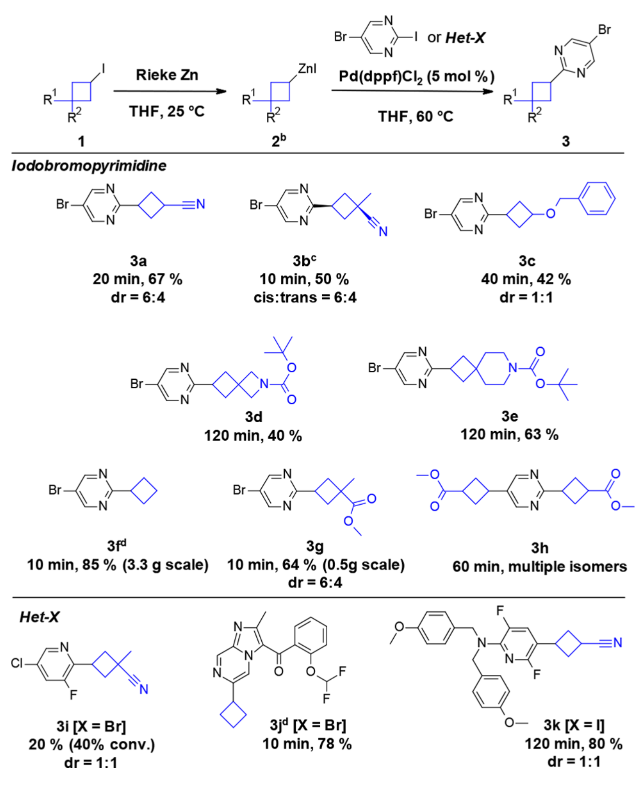
[a]Conversion determined by capillary GC-MS analysis after protic workup. [b]Zn powder dried for 5 min at 400 °C under high vacuum. [c]Rieke zinc was purchased from Rieke Metals.

of zinc metal, preactivated under Knochel's conditions (zinc washed with 1 M HCl),[12] we observed incomplete conversion with bromo and iodo derivatives (entries 3 and 4). Therefore, we performed the insertion using commercially available Rieke zinc.[13] Although the conversion was incomplete with **1a′** (entry 5), the **1a** analogue displayed full conversion after 20 min (entry 6). No preactivation of the zinc source was needed with this commercially available highly activated zinc. Alternatively, we decided to investigate the zinc insertion flowing the iodocyclobutyl derivative toward a column containing metallic zinc, previously described by Alcazar.[17] After preactivation of the zinc powder with HCl (1 M), iodo-3-cyclobutane carbonitrile **1a** (0.5 M in THF) was passed through the column at 80 °C for 6 min, and low conversion toward the organozinc reagent **2a** was observed (entry 7). Full conversion was obtained at 120 °C under 8 bar of pressure (entry 8). A concentration of 0.4 M (0.5 M expected) of the freshly prepared organozinc was observed using the Knochel procedure.[14] However, poor stability of the organozinc was observed over time (50% degradation after 1 h). Interestingly, **1a′** displayed almost no conversion under the same conditions (entry 9).

With two available procedures for the formation of organozinc reagents, we first evaluated the sequence under batch conditions. Freshly prepared (3-cyanocyclobutyl)iodozinc **2a** (1.1 equiv) was added to a mixture of the iodobromopyrimidine, Pd(dppf)-Cl_2 (5 mol %), and THF. The reaction mixture was stirred for 1 h at 60 °C. We observed the exclusive formation of the C2 monoalkylated product **3a** in 67% yield as a mixture of *cis* and *trans* isomers in a ratio of 6/4 easily separated by chromatography.

Encouraged by our preliminary results, we evaluated the scope of this transformation with various 1,3-substituted cyclobutylzinc reagents, generated using the batch zinc insertion procedure (Scheme 1). 1,3-Substituted cyclobutane **3b** was obtained in 50% yield. In this case, the *cis* isomer was determined to be the major isomer in a ratio of dr = 6:4. The *O*-benzyl-substituted cyclobutyl product **3c** was obtained in 42% yield, and no diastereoselectivity was detected. Iodocyclobutyls **1d** and **1e** with nitrogen-containing functional groups were also

Scheme 1. Scope of Pd-Catalyzed Cross-Coupling of Substituted Cyclobutyl Zinc with Aryl Halides in Batch Mode[a,e]



[a]Diastereoselectivity (*trans*/*cis*) determined by ^{1}H NMR or by capillary GC-MS analysis. [b]Complete conversion of iodocyclobutyl monitored by GC-MS analysis. [c]Major isomer displayed. [d]cBuZnBr purchased from Aldrich. [e]For compounds **3a**, **3b**, and **3g**, side products arising from the Blaise reaction were observed as minor impurities.

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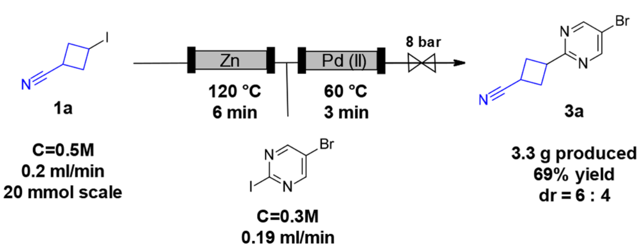
submitted to the sequence, leading to Boc-protected spirocyclobutyl derivatives **3d** and **3e** in 40 and 63% yields, respectively. Cyclobutyl iodide **1e** was also probed toward zinc insertion under flow conditions (120 °C, 6 min); however, Boc deprotection occurred, highlighting the advantage of a mild batch protocol when using thermosensitive compounds.[15] The reaction can be performed on gram scale using commercially available cyclobutylzinc bromide **2f** and the iodo(3-methoxycarbonyl-3-methylcyclobutyl)zinc **2g** and delivered the desired compounds **3f** and **3g** in good yields (85 and 64%, respectively). Surprisingly, using iodo(3-methoxycarbonylcyclobutyl)zinc **2h**, we observed bisalkylation, leading to a complex mixture of isomers **3h**. In contrast, a high level of regioselectivity[3d] was observed when 2-bromo-5-chloro-3-fluoropyridine was reacted with (3-cyano-3-methylcyclobutyl)iodozinc, despite a low conversion (20% yield, **3i**). Finally, the methodology was successfully applied to more complex bromo- and iodoheteroaryls to obtain **3j** (78%) and **3k** (80%), respectively. These last examples demonstrate the tolerance of organozinc with functional groups and the utility of this methodology to functionalize druglike molecules.

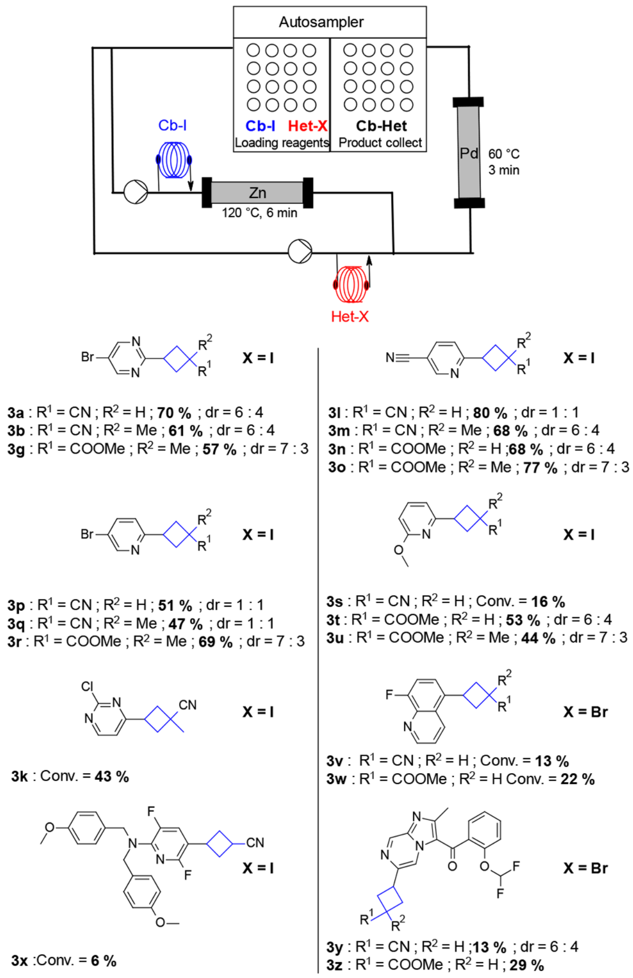
Having recognized the potential of this methodology to introduce complex cyclobutyls to heteroaromatics, we investigated the possibility of performing a flow multistep process[16] to react organozincs when they are produced and avoid handling this sensitive organometallic reagent.[17] Therefore, we combined our initial zinc insertion flow protocol (Table 1, entry 8) with a Negishi protocol in flow using a column containing 1 g of SiliaCat DPP-Pd (Scheme 2).[18] After optimization, 1.4 equiv of **1a** reagents relative to the iodobromopyrimidine is required to allow full conversion at 60 °C in THF over <3 min (Scheme 2).

Scheme 2. Pd-Catalyzed Cross-Coupling of Substituted Cyclobutyl Zinc with Iodobromopyrimidine in Flow



The flow reaction was performed on 20 mmol scale, affording 3.3 g of coupling product **3a** in 69% yield, demonstrating the stability and robustness of the process over time. To establish the scope of this flow methodology and also prepare a small library of compounds for the exploration of the new chemical space offered by this array of heteroaromatic substituted cyclobutyls (Het-Cbs), we investigated automation of the process. Integration of an autosampler to our flow system enabled us to automatically load reagents and collect products. Therefore, we submitted several iodocyclobutyls (Cb-I) and haloheteroaryls (Het-X) to the zinc insertion/Negishi coupling sequence under our optimized conditions (Scheme 3). This automated platform allowed the preparative synthesis of one coupling product (0.3 mmol scale) per hour and show efficiency over 6 h using the same column.[19] Cyclobutylbromopyrimidines **3a**, **3b**, and **3g** were obtained, affording similar yields and diastereoselectivities than in batch mode. 5-Cyano-2-iodopyridines were converted to products **3l**–**3o** in good yields (68–80%); the nitrile functionality was not affected in the transformation and offered an additional derivatization of the heteroaromatic ring. 5-Bromo-2-iodopyridines **3p**–**3r** displayed exclusive iodo-selectivity in moderate yields ranging between 47 and 69%. With these heteroaromatics, the diastereoselectivity appears to be low, with the best ratio of 7:3 obtained in both cases with **2g**. Interestingly, 2-methoxypyridine affords only 16% conversion toward CN-cyclobutylpyridine **3s**, whereas the esters analogues display full conversion and yield products **3t** and **3u** in 53 and 44% yield.

Scheme 3. Pd-Catalyzed Cross-Coupling of Substituted Cyclobutyl Zinc in Flow Using an Automated Platform[a]



[a]dr (trans/cis) determined by ^{1}H NMR [b] *c*BuZnBr purchased from Aldrich.

2-Chloro-4-iodopyrimidine, 5-bromo-8-fluoroquinoline, and protected 3,6-difluoro-5-iodopyridin-2-amine displayed incomplete conversion toward **3k** and **3v**–**3x** under standard conditions within the 3 min residence time. Those results highlight the lower reactivity of those starting materials;[20] however, extending the residence time should improve the conversions, as demonstrated by the complete conversion toward **3k** in batch mode (Scheme 1). Finally, our druglike molecule displayed incomplete conversion under standard conditions and gave moderate 13 and 29% yields. Products **3y** and **3z** further demonstrate the utility of this method for the rapid and late-stage introduction of substituted cyclobutyls.

In summary, we have described the development of a Pd-catalyzed heteroarylation of functionalized cyclobutyls. A batch mode protocol using Rieke zinc for the organozinc generation

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followed by heteroarylation allowed for the introduction of a variety of substituted cyclobutyl-bearing orthogonal functionalities. To make this sequence more attractive for library synthesis and structure—activity relationships, we developed an automated flow process avoiding handling of the organometallic reagents. In addition, we discovered that organozincs could be generated efficiently at 120 °C. The functional group tolerance and the selectivity of the organozinc reagents under cross-coupling conditions were demonstrated and allowed for subsequent transformation of our products. Finally, complementarity of the two methods has been established, with the operationally simple batch process, offering better yield when long reaction times are required, whereas the flow process offers the possibility of automatic generation, leading to diversity without the need for handling the organozinc reagent.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.orglett.8b03588.

Experimental procedure, characterization (PDF)

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Notes

The authors declare no competing financial interest.

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■ DEDICATION

This work is dedicated to Prof. Alexandre Alexakis on the occasion of his 70th birthday.

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(18) For further information about SiliaCat catalyst, visit http://www.silicycle.com (accessed September 18, 2018).

(19) Columns of reagents and catalysts displayed a decrease of efficiency after 6−8 h and required replacement.

(20) Electron-rich heteroaromatic substrates displayed no or very low reactivity under standard reaction conditions.

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